

Object Oriented Programming (PCC-CS503)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

Abstract data types and their specification. How to implement an ADT. Concrete state space, concrete invariant, abstraction function. Implementing operations, illustrated by the Text example.

Unit 2

Features of object-oriented programming. Encapsulation, object identity, polymorphism - but not inheritance.

Unit 3

Inheritance in OO design. Design patterns. Introduction and classification. The iterator pattern.

Unit 4

Model-view-controller pattern. Commands as methods and as objects. Implementing OO language features. Memory management.

Unit 5

Generic types and collections GUIs. Graphical programming with Scale and Swing. The software development process.

Object Oriented Programming Lab (PCC-CS593)

Course Details

Credit Points: 2

Laboratory Experiments

- Assignments on class, constructor, overloading, inheritance, overriding
- Assignments on wrapper class, arrays

- Assignments on developing interfaces- multiple inheritance, extending interfaces
- Assignments on creating and accessing packages
- Assignments on multithreaded programming
- Assignments on applet programming (Note: Use Java for programming)